

CHAPTER 9

Functorial Entanglement and Logical Classification

9.1 Composite Systems and Entanglement

In the Hilbert space formulation of quantum theory, *entanglement* refers to the phenomenon displayed by a *composite* quantum system according to which the behavior and properties of the latter, consisting of several subsystems, is *not reducible* to the properties of these individual subsystems. This phenomenon is manifest in the particular type of *correlations* found in the *joint probability distributions* of events associated with the subsystems constitutive of a composite quantum system. More precisely, in the Hilbert space formulation, a single quantum system is represented by a complex Hilbert space of states, whereas a composite quantum system is represented by the *tensor product* of the Hilbert spaces of the states of its subsystems over the complex numbers. In this formulation, it is the tensor product decomposition structure of the Hilbert space of states of the composite system which accounts for the appearance of entanglement phenomena among subsystems.

Let us briefly recall that a *state* of a quantum system is represented by a positive, Hermitian (self-adjoint) operator of trace 1 called a *density operator* (or statistical operator). The density operator is represented as a linear sum of orthogonal, one-dimensional projection operators:

$$\rho = \sum_i \lambda_i P_i$$

where the weight coefficients λ_i are positive numbers summing up to unity, that is $\sum_i \lambda_i = 1$. In the above linear sum, if only a single coefficient λ_i is different from zero, then the state is called a *pure state*—that is, $\rho = \lambda P$. Equivalently, a pure state can be characterized by a unit state vector Ψ belonging to the one-dimensional subspace of the Hilbert space (*ray*) into which P projects. Thus, a non-pure state, called a *mixed state*, is considered as a weighted linear sum (called a convex combination) of pure states (or one-

dimensional projection operators) where the total weight is one. Note that this is *different* from a superposition of pure states with complex coefficients, which is also a pure state according to the *superposition principle* of quantum theory. Moreover, a mixed state can be decomposed in many different ways in the form:

$$\rho = \sum_i \gamma_i \rho_i$$

where the coefficients λ_i are positive weights and the ρ_i are other density operators (which may correspond to pure states).

The particular form of the density operator expressed by $\rho = \sum_i \lambda_i P_i$ is called the *spectral decomposition* of the density operator in terms of *one-dimensional projectors*. In the Hilbert space formulation, each Hermitian operator has associated with it a *complete Boolean algebra* identified as a Boolean algebra of projection operators belonging to its spectral decomposition. Hence, given a set of observables of a quantum system, there always exists a complete Boolean algebra of projection operators, viz., a local Boolean subalgebra of the global non-Boolean event algebra of a quantum system with respect to which a local two-valued truth-functional assignment is meaningful, if and only if the given observables can be *simultaneously measurable*. As we have already explained, the possibility of local two-valued truth-functional assignments of the global non-Boolean event algebra of a quantum system points to the assumption that complete Boolean algebras play the role of *local Boolean logical frames* for *contextual* true/false value assignments. Thus, as introduced in chapters 2 and 3, an observational procedure is represented by a Boolean algebra of orthogonal projectors, where each one of these projectors corresponds to one of the possible results of the observation. For example, the probability for the result ε in state ρ is given by:

$$\mu_\varepsilon = \text{Tr}(\rho P_\varepsilon)$$

where the numbers (probabilities) μ_ε are obtained by the *trace* of the operator in the parentheses, whereas their positivity is induced by the positivity of ρ . Now, if we assume that the observed or measured values associated with the results ε are real numbers, represented by the coefficients ζ_ε , then the Hermitian operator

$$Z = \sum_\varepsilon \zeta_\varepsilon P_\varepsilon$$

is identified as the observable associated with this measurement procedure.

It is important to keep in mind that the form of decomposition of a density operator in terms of a *convex combination* of one-dimensional, orthogonal projection operators is *not unique* but *depends on*

the local Boolean logical frame of projectors associated with some observational procedure. Thus, the weight coefficients

$$\lambda_i = \text{Tr}(\rho P_i)$$

where P_i are one-dimensional projectors, are interpreted as *conditional probabilities* with respect to the local Boolean frame that P_i belongs to. Notice that since P_i may belong to different overlapping local Boolean frames, the coefficients λ_i are interpreted as conditional probabilities with respect to a Boolean localization system of compatible overlapping local Boolean frames. Thus, λ_i is the conditional probability of an equivalence class (Boolean germ) conditioned by the corresponding Boolean localization system. We conclude that the density operator of a quantum system provides a description of states in which all possible decompositions are, in a well-defined sense, *implicitly* present at once, albeit *potentially*.

As introduced in chapters 2 and 3, in a composite quantum system consisting of two subsystems, the Hilbert space of states is represented by the tensor product of the Hilbert spaces of the states of the two subsystems over the complex numbers, denoted by:

$$H = H_1 \otimes H_2$$

The tensor product of H_1 and H_2 considered as linear spaces over the complex numbers is the *unique* object in the category of complex linear spaces, which satisfies the following *universal property*: There exists a bilinear morphism $\tau: H_1 \times H_2 \rightarrow H_1 \# H_2$ such that given any linear space S and any bilinear morphism $f: H_1 \times H_2 \rightarrow S$ there exists a unique linear morphism $g: H_1 \# H_2 \rightarrow S$ such that the latter bilinear morphism admits the factorization $f = g \# \tau$. Consequently, factorization via the tensor product of two complex linear spaces H_1, H_2 , viz., $H_1 \# H_2$, should be understood as the *universal way to linearize any bilinear morphism* from $H_1 \times H_2$ to any other complex linear space S . Note that, since the tensor product construction is defined by a universal mapping property, it is unique up to unique isomorphism.

In the Hilbert space $H = H_1 \# H_2$ there exists a special class of state vectors, called *product states*, which have the product form

$$\Psi = \Psi^{[1]} \otimes \Psi^{[2]}$$

corresponding to the cartesian product of the state vectors of the two subsystems. But the tensor product Hilbert space $H = H_1 \times H_2$ additionally contains all *linear combinations* of such product states. More precisely, if we choose *orthonormal bases* $\Psi^{[1]}_a$ and $\Psi^{[2]}_{ba}$ in H_1 and H_2 correspondingly, a general vector state of the tensor product Hilbert space $H = H_1 \# H_2$ is written in the following form:

$$\Psi = \sum_{ab} \eta_{ab} \Psi^{[1]}_a \otimes \Psi^{[2]}_b$$

where the state vector Ψ represents a pure state of the total system composed of subsystems [1] and [2].

We observe that in general, a pure state of this total system is *not reduced to the product* of the state vectors of its two subsystems. Thus, it constitutes a correlated or *entangled state*, meaning that each subsystem does *not* possess a separable state within the composite system. Conceptually, this means that each one of the subsystems of the composite system does *not* have an individual, separable and definite state independent of the state of the composite system, and most significantly, this is the case *irrespective* of the spatial distance between the subsystems. This form of correlation, in other words, is *not* of any dynamical nature and constitutes a consequence of the universal property of the tensor product Hilbert space $H = H_1 \# H_2$ as a means of linearizing the bilinear morphisms from the cartesian product $H_1 \times H_2$. As introduced in chapter 2.8, quantum entanglement in this regard is well exemplified in experimental EPR correlations, and these will be further analyzed in section 9.4 of the present chapter.

The essential aspect of entanglement phenomena is that the behavior of the whole system is not reducible to the behavior of its constitutive subsystems—i.e., the whole is more than the sum of its parts; ¹ more specifically, the parts do not assume an individuation or localization independent of the whole. Thus there exists *a mutually implicative bidirectional relation* between the parts and the whole. The observant reader may have already realized that the general formulation of entanglement phenomena lies at the heart of the proposed sheaf-theoretic approach; and by contrast, the particular case of these phenomena as explicated by the nature of the tensor product Hilbert space of states $H = H_1 \# H_2$ constitutes only an *instance* of this general framework of *functorial entanglement*, which will be explained later.

At this stage, it is instructive to examine the possibility of assigning a notion of *partial state* to each of the subsystems [1] and [2], even though neither of these possesses an individual, separable state. This notion of partial state would thus encompass all the *statistical information* about H_1 that the density operator of the composite system ρ_{12} incorporates.

In order to be able to define such a notion of partial state for each of the subsystems [1] and [2] it is necessary to consider *local actions* of each one of [1] and [2]. By a local action of subsystem [1], for example, we mean a measurement which can be performed by an observable of subsystem [1]. This is an operation which is represented by Hermitian operators of the form $A^{[1]} \# 1^{[2]}$, where $A^{[1]}$ is a Hermitian operator in H_1 corresponding to the chosen observable. Notice that this is equivalent to employing a *local Boolean frame* consisting of projection operators belonging to the spectral decomposition of the chosen observable. Similarly, a local action of subsystem [2] is represented by Hermitian operators of the form $1^{[1]} \# B^{[2]}$. We stress the fact that none of the subsystems [1] and [2] have access to all of the observables (Boolean frames) of the composite system. More concretely, the algebras of observables of the subsystems can be obtained by the operation of *restriction* or *localization* of the algebra of observables of the composite system to each one of them.

Thus, the individual subsystems subsume the state ρ_{12} of the composite system *only partially*. We define the partial state of subsystem [1] as the *reduced density operator* $\rho_1 = \text{Tr}_2 \rho_{12}$ obtained by *partial tracing* over subsystem [2] (and analogously for subsystem [2]) by the following requirements:

$$\text{Tr}(\rho_1 A^{[1]}) = \text{Tr}(\rho_{12} (A^{[1]} \otimes 1^{[2]}))$$

$$\text{Tr}(\rho_2 B^{[2]}) = \text{Tr}(\rho_{12} (1^{[1]} \otimes B^{[2]}))$$

Thus, for all observables $A^{[1]}$ of subsystem [1], and all observables $B^{[2]}$ of subsystem [2], the reduced density operators ρ_1 and ρ_2 correspondingly constitute restrictions of ρ_{12} to the respective subsystems. We note that *many* different states ρ_{12} of the composite system may have the same restrictions on the algebras of observables of the two subsystems. Hence, from the point of view of subsystems [1] and [2], many different states of the composite system have *identical restrictions* to each subsystem. According to the above requirements, the reduced density operator ρ_1 , for example, reproduces the same statistical distribution for an event caused by a local action of subsystem [1] as ρ_{12} does, in the sense that we could either apply the operation $A^{[1]} \# 1^{[2]}$ on the composite system (thus leaving subsystem [2] unaffected) or apply the operation $A^{[1]}$ directly on subsystem [1]. Note that the assumption of considering an event caused by a local action of subsystem [1] leaving subsystem [2] unaffected respects the requirement of *Einstein locality* in spacetime if the two subsystems are sufficiently separated; in other words, the probability of an event at subsystem [1] is independent of subsystem [2] if the regions associated with these subsystems are *spacelike separated*.

However, it is important to realize that the reduced density operators ρ_1 and ρ_2 are *not* sufficient to determine the *probabilities of pairs* of correlated events between the two subsystems. These pairs of correlated events do not have any dynamical origin but are implied by the entanglement of the states of the composite system if we consider *compatible local actions* of the subsystems—i.e., measurements which can be performed by *compatible observables* of subsystems [1] and [2]. Equivalently, correlations between events constitutive of the subsystems can be observed with coincidence measurements performed between compatible local Boolean frames within some Boolean localization system of the composite system—i.e., Boolean frames representing the local measurement contexts of the compatible, correlated observables.

The condition of local Boolean frame compatibility between observables of the subsystems [1] and [2] means that the reduced density operators ρ_1 and ρ_2 constitute restrictions of some pure state of the composite system only if their *eigenvalues* are identical with respect to these compatible Boolean frames. Equivalently, a vector state of the tensor product Hilbert space $H = H_1 \# H_2$ reflecting the condition of local Boolean frames compatibility, is written in the following form:

$$\Psi = \sum_j \eta_j \Psi^{[1]}_j \otimes \Psi^{[2]}_j$$

where the state vector Ψ represents a pure state of the composite system; $\Psi^{[1]}_j, \Psi^{[2]}_j$ are orthonormal bases of the Hilbert spaces H_1 and H_2 of the subsystems [1] and [2], respectively, corresponding to the spectral decompositions of ρ_1 and ρ_2 with respect to compatible Boolean frames of [1] and [2] (or compatible local actions of the two subsystems); and η_j are the identical eigenvalues of [1] and [2] with respect to these bases.

In the physical state of affairs the correlated pairs of events usually refer to some *conserved* physical quantity like charge, energy, momentum or spin orientation of the subsystems (corresponding to some specified observable of the combined system) and persist *irrespective* of the metrical distance between the subsystems. It is important for the understanding of these entanglement correlations to emphasize the significance of an *event ontology*, together with the crucial role that the *compatibility between local Boolean frames* (with respect to which events occur by measurement of corresponding observables) plays, in Boolean localization systems. This is the case because entanglement correlations *cannot be reduced* to correlations between assumed preexisting states assigned to the subsystems before the

occurrence of the events (with respect to their corresponding local Boolean frames). This conclusion reaffirms the significance of Whitehead's event ontology in the relational realist understanding of entanglement correlations in quantum theory.

9.2 Functorial Entanglement

The ontological significance of events in interpreting entanglement phenomena forces a careful re-thinking of the notion of a subsystem of a composite system as exemplified by the concept of a partial state, or reduced density operator pertaining to its description. Since a partial state of a subsystem is used for the evaluation of probabilities of the events referring to this subsystem exclusively, irrespective of any correlations of these events with events referring to any other subsystem, it becomes clear that:

[i] The notion of a subsystem is *not fundamental* and retains meaning only insofar as a *complete Boolean frame* is designated. This frame corresponds to the measurement of some observable via the registration of some corresponding fact interpreted as an observed event. Hence, the notion of a subsystem before the existence of some observed events remains only as a *potentiality* expressed by its partial state description. This potentiality, under the designation of some Boolean frame, acquires the interpretation of a *probability function* for the evaluation of event-probabilities when the subsystem is definable as a reference linkage among observed events referring to the corresponding observable.

[ii] The separation of a composite system into subsystems is an *idealization*; it does not correspond to a 'partition' of the system into subsystems in terms of the density operators by which the subsystems are described. Rather, it is properly understood as the *algebraic operation of restriction or localization of the algebra of observables* of the composite system into appropriate *subalgebras* of observables corresponding to *potential subsystems* which can be realized only after the designation of local Boolean frames. Intuitively, these subalgebras represent only the observables subsumed by the potential subsystems, distinguished in this way only after the generation of concrete measurement events. Furthermore, this observables-induced distinguishability of subsystems within a composite system, for example of the subsystems [1] and [2] according to the preceding, is effectuated by considering observables of the form $A^{[1]} \# 1^{[2]}$ and $1^{[1]} \# B^{[2]}$ within the algebra of observables of the total system.

[iii] The observables-induced distinguishability of potential subsystems within a composite system allows for an understanding of entanglement correlations between these subsystems under the condition of *compatibility* between their corresponding local Boolean frames within the Boolean localization system of a composite system. For example, in the case of two potential subsystems, the condition of compatibility means that the reduced density operators ρ_1 and ρ_2 constitute restrictions of some pure

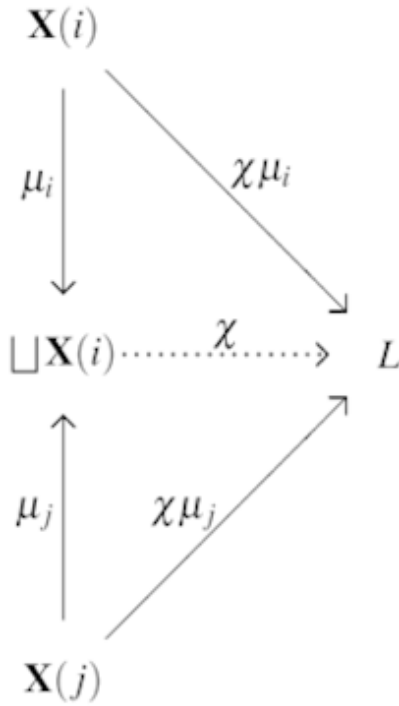
state of the composite system only if their eigenvalues are identical with respect to these compatible Boolean frames.

Reflecting on the above, we conclude that the notion of entanglement or non-separability pertaining to the description of a composite quantum system in terms of its potential parts—and the description of the parts in terms of the system as a whole—implies an *intrinsic relativity* with respect to the *compatible local Boolean frames distinguishing the subsystems* and their correspondingly compatible observables. In this way, it becomes clear that the same pure state of a total system may entail completely different entanglement correlations depending on the *choice* of different sets of local Boolean frames. Thus, it is important to formulate the notions of entanglement and non-separability *functorially*. This means that we need a functorial framework to properly express entanglement correlations among potential subsystems of a composite system which are *observable-induced* and *do not* depend on artificial choices of subsystems. For this purpose the *sheaf-theoretic uniform fibration* partitioning a general quantum event algebra into observable sorts, indexed by local Boolean frames and thus forming Boolean localization systems, is indispensable. As the discussion proceeds, we will show that the sheaf-theoretic *inductive limit* construction of a quantum event algebra with respect to some Boolean localization system can be expressed in terms of a *functorial tensor product decomposition*, providing a general *functorial formulation of entanglement phenomena* from the *sheaf-theoretic* perspective of our approach.

It was shown in chapter 8.7 that a general quantum event algebra L is represented *isomorphically* by the inductive limit taken in the category of local Boolean frames of the sheaf $\mathbf{R}_L$ with respect to a Boolean localization system:

$$L \cong \text{Colim}\{(\mathbf{R}_L, \mathbf{B}) \rightarrow \mathbf{B}\} := \text{Colim} \mathbf{R}_L = \mathbf{L}\mathbf{R}_L$$

The functorial tensor product decomposition is obtained by expressing categorically the above colimit (inductive limit) as a *coequalizer of a coproduct*. For this purpose, we consider the colimit of any functor $\mathbf{X}: I \rightarrow \mathbf{L}$ from some index category $\mathbf{I}$ to $\mathbf{L}$. Let $\mu_i: \mathbf{X}(i) \rightarrow \mathbf{L}$, $i \in I$, be the injections into the coproduct. A morphism from this coproduct, $\chi: \mathbf{L} \rightarrow \mathbf{L}$, is determined *uniquely by the set of its components* $\chi_i = \chi \mu_i$. These components χ_i form a cocone over $\mathbf{X}$ to the quantum event algebra vertex L only when for all arrows $v: i \rightarrow j$ of the index category I the following conditions are satisfied:



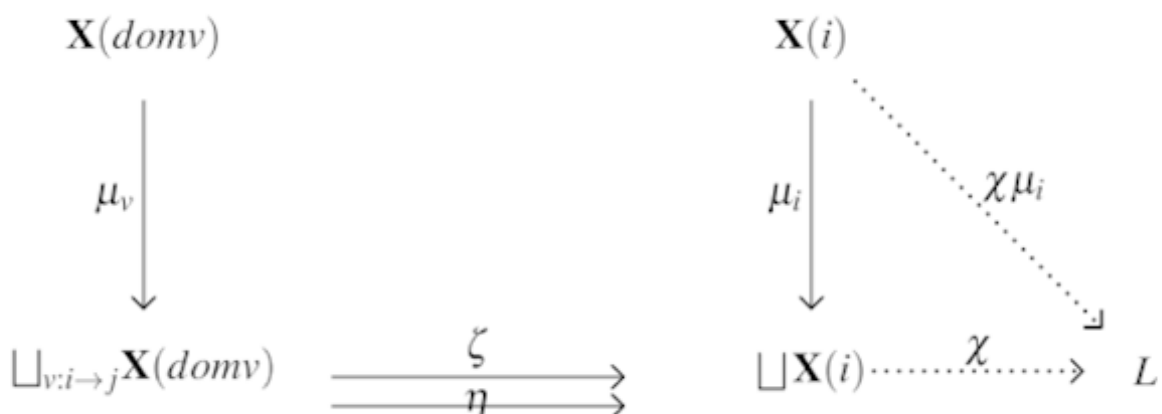
$$(\chi \mu_j) \mathbf{X}(v) = \chi \mu_i$$

So we consider all $\mathbf{X}(\text{dom}v)$ for all arrows v with its injections μ_v and obtain their coproduct $\bigsqcup_{v:i \rightarrow j} \mathbf{X}(\text{dom}v)$. Next we construct two arrows ζ and η , defined in terms of the injections μ_v and μ_i , for each $v: i \rightarrow j$ by the conditions:

$$\zeta \mu_v = \mu_i$$

$$\eta \mu_v = \mu_j \mathbf{X}(v)$$

together with their coequalizer χ :



The *coequalizer condition* $\chi\zeta = \chi\eta$ informs us that the arrows $\chi\mu_i$ form a *cocone* over $\mathbf{X}$ to the quantum event algebra vertex L . We further note that since χ is the coequalizer of the arrows ζ and η this cocone is the *colimiting cocone* for the functor $\mathbf{X}: I \rightarrow L$ from some index category I to L . Hence the *colimit of the functor $\mathbf{X}$ can be constructed as a coequalizer of coproduct* according to the following diagram:

$$\bigsqcup_{v:i \rightarrow j} \mathbf{X}(\text{dom} v) \begin{array}{c} \xrightarrow{\zeta} \\ \xrightarrow{\eta} \end{array} \bigsqcup \mathbf{X}(i) \xrightarrow{\chi} \text{Colim} \mathbf{X}$$

In our case the index category is the category of elements of the presheaf of Boolean algebras $\mathbf{P}$, that is $\int(\mathbf{P}, B)$, and the functor $\mathbf{M}\#(\mathbf{P}, B)$ plays the role of the functor $\mathbf{X}: I \rightarrow L$. In the diagram above, the second coproduct is over all the objects (B, p) with $p \in \mathbf{P}(B)$ of the category of elements, while the first coproduct is over all the maps $v:(B', p') \rightarrow (B, p)$ of that category, so that $v: B' \rightarrow B$ and the condition $p \cdot v = p'$ is satisfied. We conclude that the colimit $\mathbf{L}_M(P)$ can be equivalently presented as the following coequalizer of coproduct:

$$\bigsqcup_{v:\hat{B} \rightarrow B} \mathbf{M}(\hat{B}) \begin{array}{c} \xrightarrow{\zeta} \\ \xrightarrow{\eta} \end{array} \bigsqcup_{(B,p)} \mathbf{M}(B) \xrightarrow{\chi} \mathbf{P} \otimes_{\mathcal{B}} \mathbf{M}$$

The preceding is referred to as the *functorial tensor product decomposition* of the colimit (inductive limit) in the category of elements of $\mathbf{P}$ induced by the shaping functor of local Boolean frames $\mathbf{M}$ of a quantum categorical event structure. In order to understand the induced tensor product decomposition semantics in functorial terms, we ignore for the moment the quantum algebraic structure of the category L , and we simply take $L = \text{Sets}$. Then the coproduct $\bigsqcup_p \mathbf{M}(B)$ is a coproduct of sets, which is equivalent to the product $\mathbf{P}(B) \times \mathbf{M}(B)$ for $B \in \mathcal{B}$. The coequalizer is thus the definition of the tensor product $\mathbf{P} \#_{\mathcal{B}} \mathbf{M}$ of the set valued functors:

$$\mathbf{P}: \mathcal{B}^{op} \rightarrow \text{Sets}, \quad \mathbf{M}: \mathcal{B} \rightarrow \text{Sets}$$

$$\bigsqcup_{B, \hat{B}} \mathbf{P}(B) \times \text{Hom}(\hat{B}, B) \times \mathbf{M}(\hat{B}) \begin{array}{c} \xrightarrow{\zeta} \\ \xrightarrow{\eta} \end{array} \bigsqcup_B \mathbf{P}(B) \times \mathbf{M}(B) \xrightarrow{\chi} \mathbf{P} \otimes_{\mathcal{B}} \mathbf{M}$$

According to the above diagram for elements $p \in \mathbf{P}(B)$, $v: B' \rightarrow B$ and $q' \in \mathbf{M}(B')$ the following equations hold, symmetric in $\mathbf{P}$ and $\mathbf{M}$:

$$\zeta(p, v, q') = (p \cdot v, q'), \quad \eta(p, v, q') = (p, v(q'))$$

Hence the elements of the set $\mathbf{P} \#_{\mathbf{B}} \mathbf{M}$ are all of the form $\chi(p, q)$, which can be equivalently written as:

$$\chi(p, q) = p \otimes q, \quad p \in \mathbf{P}(B), q \in \mathbf{M}(B)$$

Thus if we take into account the definitions of ζ and η above, we obtain:

$$p \cdot v \otimes q' = p \otimes v(q'), \quad p \in \mathbf{P}(B), q' \in \mathbf{M}(B'), v: B' \rightarrow B$$

We conclude that the set $\mathbf{P} \#_{\mathbf{B}} \mathbf{M}$ is actually the *quotient* of the set $\sqcup_B \mathbf{P}(B) \times \mathbf{M}(B)$ by the *equivalence relation* generated by the above equations. Furthermore if we define the arrows

$$k_B: \mathbf{P} \otimes_{\mathbf{B}} \mathbf{M} \rightarrow L, \quad l_B: \mathbf{P}(B) \rightarrow \text{Hom}_L(\mathbf{M}(B), L)$$

they are related under the Boolean frames-quantum adjunction by:

$$k_B(p, q) = l_B(p)(q), \quad B \in \mathbf{B}, p \in \mathbf{P}(B), q \in \mathbf{M}(B)$$

Here we consider k as a function on $\sqcup_B \mathbf{P}(B) \times \mathbf{M}(B)$ with components $k_B: \mathbf{P}(B) \times \mathbf{M}(B) \rightarrow L$ satisfying

$$k_{B'}(p \cdot v, q') = k_B(p, v(q'))$$

in agreement with the equivalence relation defined above.

Now we replace the category *Sets* by the category of quantum event algebras L under study. The element q in the set $\mathbf{M}(B)$ is replaced by a generalized element $q: C \rightarrow \mathbf{M}B$ from some object C of L . Then we consider k as a function $\sqcup_{B, p} \mathbf{M}(B) \rightarrow L$ with components $k_{(B, p)}: \mathbf{M}(B) \rightarrow L$ for each $p \in \mathbf{P}(B)$, which for all arrows $v: B' \rightarrow B$ satisfy:

$$k_{(B', p \cdot v)} = k_{(B, p)} \circ \mathbf{M}(v)$$

Then the condition defining the bijection holding by virtue of the Boolean frames-quantum adjunction is given by:

$$k_{(B, p)} \circ q = l_B(p) \circ q: C \rightarrow L$$

This argument, being natural in the object C , is determined by setting $C = \mathbf{M}(B)$ with q being the identity map. Hence the bijection takes the form $k_{(B, p)} = l_B(p)$.

In particular, if $\mathbf{P}$ is the sheaf of local Boolean frames $\mathbf{R}_L$ with respect to a Boolean localization system of L , such that locally:

$$L \cong \text{Colim}\{\mathbf{R}_L \rightarrow \mathbf{B}\} := \text{Colim} \mathbf{R}_L$$

then we obtain:

$$\bigsqcup_{v: B' \rightarrow B} \mathbf{M}(B') \xrightarrow[\eta]{\zeta} \bigsqcup_{(B, \psi_B)} \mathbf{M}(B) \xrightarrow{\chi} \mathbf{R}(L) \otimes_{\mathcal{B}} \mathbf{M}$$

According to the diagram above for elements $\Psi_B \in \mathbf{R}(L)(B)$, $v: B' \rightarrow B$ and $\xi' \in \mathbf{M}(B')$, the following equations hold:

$$\zeta(\psi_B, v, \xi') = (\psi_B \cdot v, \xi'), \quad \eta(\psi_B, v, \xi') = (\psi_B, v(\xi'))$$

symmetric in $\mathbf{R}(L)$ and $\mathbf{M}$. Hence the elements of $\mathbf{R}(L) \#_B \mathbf{M}$ are all of the form $\chi(\Psi_B, \xi)$. This element can be written as:

$$\chi(\psi_B, \xi) = \psi_B \otimes \xi, \quad \psi_B \in \mathbf{R}(L)(B), \xi \in \mathbf{M}(B)$$

Thus if we take into account the definitions of ζ and η above, we obtain:

$$\psi_B \cdot v \otimes \xi' = \psi_B \otimes v(\xi')$$

We conclude that $\mathbf{R}(L) \#_B \mathbf{M}$ is actually the quotient of $\bigsqcup_{(B, \psi_B)} \mathbf{M}(B)$ by the smallest equivalence relation generated by the above equations. Moreover, if there exists a Boolean algebra D in $\mathcal{B}$ and Boolean homomorphisms $w: D \rightarrow B$, $h: D \rightarrow B'$, such that:

$$w(\bar{\xi}) = \xi, h(\bar{\xi}') = \xi', \psi_B \cdot w = \psi_D, \psi_{B'} \cdot h = \psi_D \text{ and thus } \psi_B \cdot w = \psi_{B'} \cdot h$$

where

$$\xi \in \mathbf{M}(B), \xi' \in \mathbf{M}(B'), \bar{\xi} \in \mathbf{M}(D), \psi_B \in \mathbf{R}(L)(B), \psi_{B'} \in \mathbf{R}(L)(B'), \psi_D \in \mathbf{R}(L)(D)$$

then the identification equations take the general symmetric form:

$$\psi_B \otimes \xi = \psi_{B'} \otimes \xi'$$

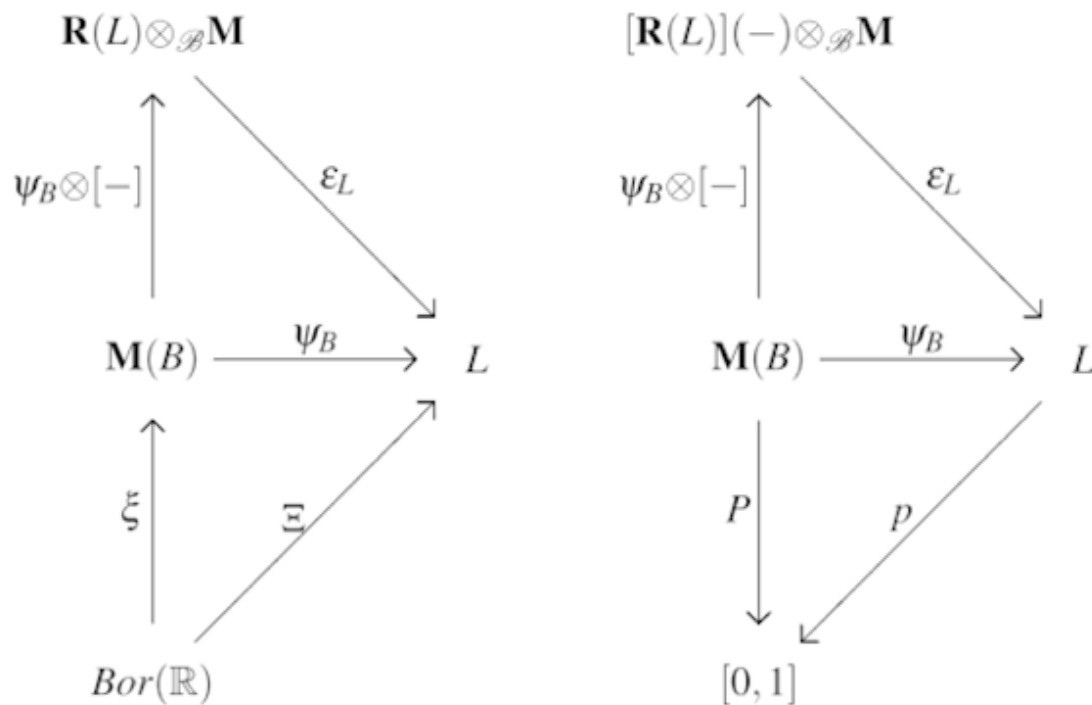
If we denote by $l_Q(B)$ the ideal generated by the equivalence relation, corresponding to the above identification equations, for each B in $\mathcal{B}$, we conclude that locally—i.e., with respect to a Boolean localization system—an element of $\mathbf{R}(L) \#_{\mathcal{B}} \mathbf{M}$ can be written in the form of a *Boolean germ*:

$$\psi_B \otimes \xi = (\psi_B, \xi) + l_Q(B)$$

Thus, a *quantum event algebra* admits a *sheaf-theoretic representation* so that it can be described *locally*—that is, over a particular cover in a Boolean localization system, as follows:

$$\mathbf{R}(L)(B) \otimes_{\mathcal{B}} \mathbf{M}(B) = [\mathbf{R}(L)(B) \times \mathbf{M}(B)] \text{ modulo } [l_Q(B)]$$

Furthermore, we mention that precisely analogous conclusions hold if we consider the categories of quantum observables and quantum probabilistic states defined over a quantum event algebra according to the following commutative diagrams:



9.3 Interpretation of Quantum Events

If we focus our attention on a Boolean localization system of compatible overlapping local Boolean frames, we can accurately specify the information encoded in the *ideal* $l_Q(B)$ in $\mathbf{R}(L)(B) \times_{\mathcal{B}} \mathbf{M}(B)$, where $\Psi_B : \mathbf{M}(B) \rightarrow L$, and B in $\mathcal{B}$, stands for a local Boolean frame belonging to this system. More concretely, the ideal $l_Q(B)$ contains *information about all the other local Boolean frames in the*

localization system which are compatible in pullback diagrams over $L \# \mathbf{R}(L) \#_{\mathbf{B}} \mathbf{M}$ with the specified one. This is clear if we inspect the gluing isomorphism $\Omega_{B,B'} = \Psi_{BB'} \# \Omega_{B'B}^{-1}$ and notice that its existence guarantees the satisfaction of the relations needed for the validity of the identification equations: $\Psi_B \times \xi = \psi_{B'} \# \xi$ in a Boolean localization system. Thus, the information encoded in the ideal $I_Q(B)$ essentially refers to all other local Boolean frames that are compatible with some specified one in a Boolean localization system. This is a *uniquely peculiar characteristic of the local behavior of quantum event algebras in comparison to classical event algebras*. Remarkably, each observable-induced local Boolean frame in a covering sieve (Boolean localization system) of a global quantum event algebra incorporates local observable information (in the form of a *Boolean germ*) about *all other local Boolean frames compatible with it in the same covering sieve*. This is a crucial observation and pertains to discussions of alleged ‘quantum non-local causality’ “characterizing the supposedly ‘paradoxical’ behavior of quantum systems. As discussed earlier in chapters 2 and 3, from our perspective, the improper characterization of non-local correlations as ‘paradoxical’ stems from two factors: The first factor has to do with the failure to acknowledge that entanglement correlations between potential subsystems are *observable or Boolean-frame induced, presupposing the existence of events*. The second factor stems from the misconceived attempt to *assimilate the potential relations, described via topologically interlocking local Boolean frames constitutive of a Boolean localization system, to actual causal relations*—i.e., in terms of a non-local, metrical, efficient causal mechanism of information propagation between spacetime points.

The *Boolean germ relation*—that is, the equivalence relation of local Boolean frames belonging to a Boolean localization system of a quantum event algebra—provides the following insights for the interpretation of events in a quantum world:

The first is that the Boolean germ relation, which is observable induced, establishes a *criterion of equivalence* among experimental measurement contexts / observational procedures. Thus, in a Boolean localization system of L using local Boolean frames $(B, \psi_B : \mathbf{M}(B) \rightarrow L)$ and Boolean coefficients $b \in B$ we can form their equivalence classes, which modulo the compatibility conditions on overlaps, will represent a *single quantum event* in L . The fact that two different local Boolean frames $(B, \psi_B : \mathbf{M}(B) \rightarrow L)$ and $(C, \psi_C : \mathbf{M}(C) \rightarrow L)$ in a Boolean localization system are compatible, reflects the possibility of using *two distinct experimental contexts* of compatible observables, or equivalently setting up two different measurement procedures of compatible observables, and hence obtaining two outcome events quantified by the Boolean propositions $b \in B$ and $c \in C$. Now, the equivalence relation between these two local Boolean frames means that the corresponding observational procedures are in fact *equivalent*, and thus they record the same quantum event.

The second insight is provided by the application of the *Stone representation theorem* for Boolean algebras.² This theorem establishes a *duality* (categorical equivalence) between the category of *Boolean event algebras* and the category of *compact totally disconnected Hausdorff topological spaces* (Stone spaces). In the finite case this duality restricts to an equivalence of the category of finite Boolean algebras and the opposite of the category of finite sets. In this way, each Boolean algebra B has an associated topological space $S(B)$, called its *Stone space*. The points of the Stone space $S(B)$ are the *ultrafilters* of the Boolean algebra B , or equivalently the homomorphisms from B to the 2-element Boolean algebra. A basis of the topology of $S(B)$ consists of sets of the form $\{x \in S(B) \mid b \in x\}$ where $b \in B$. Conversely, given any topological space X , the collection of subsets of X that are compact open or equivalently clopen, that is, both closed and open, is a Boolean algebra.

Now, the assignment of a Boolean algebra to its Stone space is *functorial*, that is, it corresponds to the object part of a contravariant functor from the category of Boolean algebras to the category of Stone spaces

$$Hom_B(-, 2) : B^{op} \rightarrow T$$

where B, T are the categories of Boolean algebras and Stone spaces correspondingly. The functor $Hom_B(-, 2)$ sends a Boolean algebra B to the Stone space $S(B) := X_B$, and a Boolean algebra homomorphism $F : B \rightarrow A$ to the continuous function $Hom_B(F, 2)$, whose action is defined by precomposition with F , viz., it sends $H : A \rightarrow 2$ of X_A to $F \# H : B \rightarrow 2$ of X_B . Now, there exists an inverse functor

$$Hom_T(-, 2) : T \rightarrow B^{op}$$

which sends a Stone space S to the set of continuous functions from S to the Stone space 2 , endowed with the structure of a Boolean algebra, and further identified as the Boolean algebra of its clopen sets. Moreover, it sends a continuous function of Stone spaces $f : S_1 \rightarrow S_2$ to the homomorphism of Boolean algebras $Hom_T(f, 2)$, whose action is defined by postcomposition with f correspondingly. Notice that the *duality* between the category of Boolean algebras and the category of Stone spaces has been based on a *dual interpretation of the two-element set 2*, viz., as the two-element true/false Boolean algebra or as a two-point Stone space.

According to the above duality, it is legitimate to replace Boolean algebras by *fields of subsets of a sample space*, obtaining in physical grounds an operational instantiation of the notion of observed event. Hence, if we replace each Boolean algebra B in B by its set-theoretic representation $[\Sigma, B_\Sigma]$, consisting of a local sample space Σ and its local field of subsets B_Σ , it is possible to define local sample space

covers $(B_\Sigma, \psi_{B_\Sigma}: \mathbf{M}(B_\Sigma) \rightarrow L)$ and corresponding localization systems for a quantum event algebra L in L . Again from local sample space covers $(B_\Sigma, \psi_{B_\Sigma}: \mathbf{M}(B_\Sigma) \rightarrow L)$ we may form their *equivalence classes*, which, modulo the conditions for compatibility on overlaps, will represent a single quantum event in L . Under these circumstances we may interpret the equivalence classes of local sample space covers ψ_{B_Σ} , $\# a, a \in \mathbf{M}(B_\Sigma)$ as the *statistical experimental actualizations* of the quantum events in L . Moreover, the local sample space covers $(B_\Sigma, \psi_{B_\Sigma}: \mathbf{M}(B_\Sigma) \rightarrow L)$ and $(C_\Sigma, \psi_{C_\Sigma}: \mathbf{M}(C_\Sigma) \rightarrow L)$ are compatible in a localization system for L , if and only if for some $(D_\Sigma, \psi_{D_\Sigma}: \mathbf{M}(D_\Sigma) \rightarrow L)$ in the localization system, and $a \in \mathbf{M}(B_\Sigma), b \in \mathbf{M}(C_\Sigma), c, d \in \mathbf{M}(D_\Sigma)$ the following conditions hold:

$$\psi_{B_\Sigma} \otimes a = \psi_{D_\Sigma} \otimes c$$

$$\psi_{C_\Sigma} \otimes b = \psi_{D_\Sigma} \otimes d$$

The pullback compatibility condition (topological gluing isomorphism), which is in injective correspondence with the one in L since it holds in a localization system, is interpreted in the measurement context as denoting that two local sample space representations of a quantum event are compatible if and only if their corresponding measurements are equivalent to measurements conducted in experimental contexts of compatible observables. Thus, the equivalence relation between these two local sample spaces means that the corresponding observational procedures are in fact *equivalent*, and thus they record the same quantum event.

We conclude that it is precisely this internal equivalence relation between compatible observable-induced Boolean frames in a localization system, obtained by the functorial tensor product representation of the inductive limit (in the category of local Boolean frames of the sheaf $\mathbf{R}_L$), which permits an indirect logical inference from one Boolean frame to another after some event has actualized with respect to one of them.

9.4 EPR Correlations

As a particular application of the functorial scheme of sheaf-theoretic entanglement we have developed so far, it is instructive to include a brief discussion of *EPR correlations*, which were introduced in chapter 2.8. The EPR correlations are due to the phenomenon of *entanglement* or *non-separability* pertaining to the description of a quantum system with reference to *two potential subsystems* emerging by the depiction of certain compatible local Boolean frames distinguishing them and

corresponding to compatible observables. For example, in Bohm's version of the EPR correlations, these refer to correlated pairs of events associated with two compatible observable-induced distinguished subsystems (*spin in some designated direction*) and are due to *total spin conservation*, persisting *irrespective* of the metrical distance between the subsystems.

The hallmark of EPR correlations is that they *cannot* be reduced to correlations between assumed, pre-existing states assigned to the subsystems before the occurrence of measurement events (with respect to their corresponding compatible observable-induced local Boolean frames). Thus, our understanding of EPR correlations is based on an *event ontology* and their explanation stems from the *sheaf-theoretic compatibility relation* between local Boolean frames (with respect to which events occur by measurement of corresponding compatible observables) in Boolean localization systems. These Boolean localization systems are *indexed by the axis of the observable spin*, since spin observables having different spin axes are not compatible.

Notice that the two *observable-induced subsystems* constitutive of an EPR-pair *do not* assume an individuation or localization independent of the total system. More precisely, for example, the *singlet pure state* of the total system is not reducible to a product which would be assigned to the state vectors of the two subsystems. Thus, it is an entangled state, which means that each subsystem does not possess a separable state independent of the state of the total system. Thus, there exists a *mutually implicative bidirectional relation* between the observable-induced subsystems and the total system. This relation is manifest via *correlation events* generated by coincidence spin measurements in designated directions (induced by the actions of corresponding detectors):

$$\Psi = (1 / \sqrt{2}) \{ \Psi^{[1]}_{[+]} \otimes \Psi^{[2]}_{[-]} - \Psi^{[1]}_{[-]} \otimes \Psi^{[2]}_{[+]} \}$$

A natural question is whether it is possible to write the singlet state above as a kind of *composition of 'potential parts'* which become *observable-induced*, thus reflecting their *non-separability*. We denote the 'potential parts' by $\widehat{\Psi}_1$ and $\widehat{\Psi}_2$ correspondingly:

$$\widehat{\Psi}_1 = (1 / \sqrt{2}) \{ \widehat{\Psi^{[1]}_{[+]}} \otimes \Psi^{[2]}_{[-]} - \widehat{\Psi^{[1]}_{[-]}} \otimes \Psi^{[2]}_{[+]} \}$$

$$\widehat{\Psi}_2 = (1 / \sqrt{2}) \{ \Psi^{[1]}_{[+]} \otimes \widehat{\Psi^{[2]}_{[-]}} - \Psi^{[1]}_{[-]} \otimes \widehat{\Psi^{[2]}_{[+]}} \}$$

$$\Psi = \widehat{\Psi}_1 \mathbf{e} \widehat{\Psi}_2$$

Notice that the ‘potential parts’ $\overline{\Psi_1}$ and $\overline{\Psi_2}$ are *not-separable* from the entangled singlet state Ψ , since they *do not* constitute tensor factors of Ψ . As we have shown, the notion of ‘potential parts’ in our scheme, to be interpreted as states of potential subsystems, emerges by the operation of restriction or localization of the algebra of observables of the total system to these subsystems. Now, the *observable-induced distinguishability* of these potential subsystems by events pertaining to $\overline{\Psi_1}$ or $\overline{\Psi_2}$, for instance measurement of spin in some direction, allows an understanding of correlated pairs of events between the subsystems under the *condition of compatibility* between their corresponding local Boolean frames (i. e., spin measurement in the same direction) within a Boolean localization system of a composite system (indexed by the direction of spin measurement). The *condition of compatibility* means that given the reduced density operators ρ_1 and ρ_2 of the potential subsystems, they constitute *restrictions of the singlet state* of the total system, and thus pertain to $\overline{\Psi_1}$ and $\overline{\Psi_2}$, only if their eigenvalues are identical (*perfectly anti-correlated*) with respect to these compatible Boolean frames of spin measurement in a designated direction as an implementation of the *total spin conservation* principle. Notice that the singlet state is characterized by *rotational invariance* so the above argument is valid *independent* of the choice of direction of spin measurement *modulo* the compatibility of the local Boolean frames of the two subsystems (corresponding to the spin observable in each one of them in the designated direction).

For example, let us consider the spin observable in the z-direction $S_z^{[1]}$ pertaining to the ‘potential part’ $\overline{\Psi_1}$, and thus indexing it by the direction z (we note that the spin operators can be represented using the standard Pauli matrices). The measurement of the observable $S_z^{[1]}$ induces a local action of subsystem [1] (distinguished by the subalgebra of observables of the total system ‘visible’ to it), which under the assumption that it leaves subsystem [2] unaffected (and thus respects Einstein locality) has the form $S_z^{[1]} \# 1^{[2]}$. This is equivalent to employing a *local Boolean frame* consisting of projection operators belonging to the spectral decomposition of $S_z^{[1]} \# 1^{[2]}$, which we denote by B . Analogously, we may consider the spin observable in the z-direction $S_z^{[2]}$ pertaining to the ‘potential part’ $\overline{\Psi_2}$ and thus indexing it by z. Symmetrically, the measurement of the observable $S_z^{[2]}$ induces a local action of subsystem [2] of the form $1^{[1]} \# S_z^{[2]}$, which is equivalent to employing a corresponding local Boolean frame, denoted by B' . Then, we define the *partial state* of subsystem [1] induced by the local action $S_z^{[1]}$ of subsystem [1], as the reduced density operator $\rho_1 = Tr_{[2]} \rho_{12}$ (where ρ_{12} denotes the density operator of the total system corresponding to the singlet state) obtained by partial tracing over subsystem [2] (and analogously for subsystem [2]) by:

$$Tr(\rho_1 \mathbf{S}_z^{[1]}) = Tr(\rho_{12} (\mathbf{S}_z^{[1]} \otimes 1^{[2]}))$$

$$Tr(\rho_2 \mathbf{S}_z^{[2]}) = Tr(\rho_{12} (1^{[1]} \otimes \mathbf{S}_z^{[2]}))$$

Thus, we obtain:

$$\rho_1 = (1/2)(\Pi^z_{\Psi^{[1]}_{[+]}} + \Pi^z_{\Psi^{[1]}_{[-]}})$$

$$\rho_2 = (1/2)(\Pi^z_{\Psi^{[2]}_{[+]}} + \Pi^z_{\Psi^{[2]}_{[-]}})$$

where $\Pi^z_{\Psi^{[1]}_{[+]}}$ is the *projection operator* of the local Boolean frame in the direction z for subsystem [1], which projects in the $[+]$ orientation and similarly for the others.

We note that many different density operators of the total system of the form ρ_{12} may have the same restrictions on each one of the two subsystems. Hence, from the point of view of subsystems [1] and [2], many different states of the total system have identical restrictions to each one of them and it is *not* possible to be distinguished by them. For example the state of the total system:

$$\Psi = (1/\sqrt{2})\{\Psi^{[1]}_{[+]} \otimes \Psi^{[2]}_{[-]} + \Psi^{[1]}_{[-]} \otimes \Psi^{[2]}_{[+]}\}$$

gives exactly the same partial states ρ_1 and ρ_2 induced by corresponding observables as the singlet state does. This is to be expected since the reduced density operator ρ_1 , for example, gives the statistical distribution for an event referring to a measurement of $\mathbf{S}_z^{[1]}$ leaving subsystem [2] unaffected. However, the reduced density operators ρ_1 and ρ_2 are not sufficient to determine the probabilities of pairs of correlated events between the two subsystems. These pairs of events can be observed only by coincidence measurements performed between compatible local Boolean frames, i.e., measurement of spin in z direction for subsystems [1] and [2] within some Boolean localization system of the total system corresponding to these compatible measurements. The compatibility condition with respect to local Boolean frames [1] and [2] means that the reduced density operators ρ_1 and ρ_2 constitute restrictions of the singlet pure state of the total system, characterized by total spin zero in direction z , only if their eigenvalues are identical (equal with $1/2$) with respect to these compatible Boolean frames, such that in the direction z the total spin orientation *conservation* is satisfied (perfect anti-correlation).

Taking into account our previous analysis we realize that there exists an *internal equivalence relation* between the compatible observables induced by $\mathbf{S}_z^{[1]} \# 1^{[2]}$ and $1^{[1]} \# \mathbf{S}_z^{[2]}$ for the Boolean frames of subsystems [1] and [2], correspondingly, in the Boolean localization system formed in the z direction for

the total system. This is obtained by the *functorial tensor product representation* of the inductive limit (in the category of local Boolean frames of the sheaf $\mathbf{R}_L$ localized in the z direction, where L is the quantum event algebra of the total system) which permits an *indirect logical inference* from one Boolean frame B to the other compatible one B' after some detection event has been registered with respect to one of them. These Boolean frames are thus *topologically amalgamated together* such that in one equivalence class (Boolean germ) of the *inductive limit* the projector $\prod_Y^Z[i]_{[+]}$ of the Boolean frame B is glued together with the projector $\prod_Y^Z[p]_H$ of B' and in another the projector $\prod_Y^Z[i]_H$ of B is glued together with the projector $\prod_Y^Z[2]_{[+]}$ of B' . The Boolean localization system in the z direction is depicted in the following diagram:

$$\begin{array}{ccc}
 T & & \\
 \swarrow u & \searrow h & \\
 & \mathbf{M}(B) \cap \mathbf{M}(B') & \xrightarrow{\psi_{B,B'}} \mathbf{M}(B) \\
 \searrow g & \downarrow \psi_{B',B} & \downarrow \psi_B \\
 & \mathbf{M}(B') & \xrightarrow{\psi_{B'}} \mathbf{R}(L) \otimes_{\mathcal{B}} \mathbf{M}
 \end{array}$$

Where this commutative diagram is considered in L , $\mathbf{M}(B)$ and $\mathbf{M}(B')$ are the images of the Boolean frames B and B' in L (the quantum event algebra of the total system) under the action of the shaping functor $\mathbf{M}$; and $\mathbf{R}(L) \#_{\mathcal{B}} \mathbf{M}$ is the inductive limit consisting of Boolean germs in the sheaf of Boolean frames $\mathbf{R}(L)$, being locally (with respect to this Boolean localization system) isomorphic to L . The gluing morphism of compatible Boolean frames on their overlap is given by:

$$\Omega_{B,B'} = \psi_{BB'} \circ \psi_{B'B}^{-1}$$

The corresponding Boolean germs are defined by the equivalence relation generated by the following identification equations:

$$\psi_B \otimes \Pi^z_{\Psi[1]_{[+]}} = \psi_{B'} \otimes \Pi^z_{\Psi[2]_{[-]}}$$

$$\psi_B \otimes \Pi^z_{\Psi[1]_{[-]}} = \psi_{B'} \otimes \Pi^z_{\Psi[2]_{[+]}}$$

If we denote by l_Q the *ideal* generated by the equivalence relation, creating *Boolean germs* as above, we conclude that locally—that is, with respect to this Boolean localization system—the quantum event algebra L over each of the Boolean frames B and B' of this system takes the form:

$$Q(B) := \mathbf{R}(L)(B) \otimes_{\mathbf{B}} \mathbf{M}(B) = [\mathbf{R}(L)(B) \times \mathbf{M}(B)] \text{ modulo } [l_Q(B)]$$

and similarly over $\mathbf{M}(B')$. Notice that the ideal $l_Q(B)$, referring to subsystem [1] from the perspective of the Boolean localization system of the total system, contains potential information about the other compatible local Boolean frame of subsystem [2] from the particular way that these Boolean frames are glued together in this Boolean localization system. Thus, it is precisely $Q(B)$, the quantum event algebra restricted to subsystem [1] from the *global* perspective of the total system, which *explains* the entanglement EPR correlations implicated in the singlet state of total spin zero of the total system—or equivalently, of the total system's 'potential part' $\overline{\Psi_1}$ (and likewise for subsystem [2]) together with the associated quantum event semantics.

9.5 Logical Classification of Quantum Events

As a logical state of affairs, observed events in quantum theory correspond to evaluations of observables with respect to some *Boolean logical binary schemata* (*yes/no experiments* subsumed by measuring devices), interpreted as *property-indicating facts*. Note that this is meaningful from a topological perspective only if it respects the normative constraints of the *relational decomposition* of a quantum event structure with respect to local Boolean frames, as well as the rules of compatibility under relational extension from the local to the global and conversely. The functionality of local Boolean frames is crucial for a relational realist interpretation of quantum theory. For this purpose it is necessary to study their functional role with respect to the logical structure and coherence of quantum theory.

We have already deduced that the representation of a quantum event algebra L in $\mathbf{L}$, in terms of the functor of Boolean frames $\mathbf{R}(L)$ of L , is full and faithful, if the Boolean frames-quantum adjunction is restricted to a system of Boolean localizations of L . As a corollary we have obtained that a quantum categorical event structure $\mathbf{L}$ is a reflection of the topos of presheaves $\mathbf{Sets}^{B^{op}}$. Thus, there exists a

subobject functor for a quantum categorical event structure L equipped with Boolean localization systems.

The *subobject functor* of the category of quantum event algebras L is defined as follows:

$$\mathbf{Sub} : L^{op} \rightarrow Sets$$

The subobject functor of L is a contravariant functor by pulling back. Composition of this functor with a proper Boolean modeling functor defines a presheaf functor in $Sets^{B^{op}}$, called the *Boolean frames modeled subobject functor* of L as follows:

$$\mathbf{Sub} \circ \mathbf{M} : B^{op} \rightarrow L^{op} \rightarrow Sets$$

In a compact notation we obtain:

$$\Upsilon_M := \Upsilon(\mathbf{M}(-)) := \mathbf{Sub} \circ \mathbf{M} : B^{op} \rightarrow Sets$$

such that:

$$B^{op} \ni B \mapsto \{[Dom(m) \circ \mathbf{M}(B)]\} \in Sets$$

where the range denotes the set of subobjects of $\mathbf{M}(B)$, viz., the set of equivalence classes of monic quantum homomorphisms (quantum monomorphisms) m from $Dom(m)$ to $\mathbf{M}(B)$.

The set $\#_M(B) = \#(\mathbf{M}(B))$ is defined as the *set of all subobjects* of $\mathbf{M}(B)$, for every B in B , in the category L . Notice that the set $\#(\mathbf{M}(B))$, for every B in B , is a *partially ordered* set under inclusion of subobjects of $\mathbf{M}(B)$.

A natural question arising in this categorical setting is the following: *Is the subobject functor representable in L by means of a concrete quantum event algebra Ω in L , which would play the role of a classifying object in L ?* The representation of the subobject functor in a quantum categorical event structure L via $\#_M$ is significant because it allows one to interpret the concrete classifying object Ω as an object that contains the *information of equivalence classes of truth valuations* with respect to all compatible Boolean frames belonging to a Boolean localization system of a quantum event algebra L in L . In that case L is endowed with a *subobject classifier*, defined as a universal monic quantum homomorphism,

$$\top := True : 1 \circ \Omega$$

such that every subobject of L in $\mathbf{L}$, is uniquely a pullback of the universal monic T , according to the following diagram:

$$\begin{array}{ccc} K & \xrightarrow{!} & 1 \\ \downarrow m & & \downarrow T \\ L & \xrightarrow{\phi_m} & \Omega \end{array}$$

We can prove the following proposition: [3](#)

The functor $\#_{\mathbf{M}}$ is representable in $\mathbf{L}$ if and only if the evaluation of the unit of the Boolean frames-quantum adjunction at $\#_{\mathbf{M}}$, restricted to a functor of Boolean localizations of L for every L in $\mathbf{L}$ is an isomorphism.

The basic idea of the proof is the following: The counit of the Boolean frames-quantum adjunction, for each quantum event algebra L in $\mathbf{L}$, is given by:

$$\varepsilon_L : \mathbf{LR}(L) \rightarrow L$$

The counit evaluated at L , that is ε_L , is a quantum algebraic isomorphism if it is restricted to a Boolean localization system of L . On the other hand, we have seen that for any presheaf $\mathbf{P} \in \mathbf{Sets}^{Bop}$, the unit is defined as follows:

$$\delta_{\mathbf{P}} : \mathbf{P} \rightarrow \mathbf{RLP}$$

It is easy to see that if we consider the Boolean frames modeled subobject functor $\#(\mathbf{M}(\#))$ as $\mathbf{P} \in \mathbf{Sets}^{Bop}$, we obtain the following natural transformation:

$$\delta_{Y(\mathbf{M}(-))} : Y(\mathbf{M}(-)) \rightarrow \mathbf{RL}Y(\mathbf{M}(-))$$

or equivalently:

$$\delta_{Y(\mathbf{M}(-))} : Y(\mathbf{M}(-)) \rightarrow \mathbf{Hom}_{\mathbf{L}}(\mathbf{M}(-), \mathbf{LY}(\mathbf{M}(-)))$$

Hence, by inspecting the unit of the adjunction arrow $\delta_{\#(\mathbf{M}(\#))}$ evaluated at $\#(\mathbf{M}(\#))$, we conclude that the *Boolean frames modeled subobject functor becomes representable* in $\mathbf{L}$ if and only if, given that the counit ε_L for every L in $\mathbf{L}$ is an isomorphism, the unit $\delta_{\#(\mathbf{B}(\#))}$ is also an isomorphism. Thus, $\#_M$ becomes representable in $\mathbf{L}$ if and only if the unit $\delta_{\#(\mathbf{M}(\#))}$, restricted to a functor of Boolean localizations of L , called the *localized unit* at $\#_M$, is an isomorphism.

As a corollary we deduce that if the evaluation of the localized unit at $\#_M$ is an isomorphism, then the *quantum classifying event algebra* is given by the inductive limit taken in the category of elements of $\#_M$, according to:

$$\Omega := \mathbf{L}\Upsilon(\mathbf{M}(-)) = \text{Colim}\{ \int(\Upsilon(\mathbf{M}(-)), \mathbf{B}) \rightarrow \mathbf{B} \rightarrow \mathbf{L} \}$$

Hence, the following diagram is a classifying pullback square in $\mathbf{L}$:

$$\begin{array}{ccc} \text{Dom}(\lambda) & \xrightarrow{\quad ! \quad} & 1 \\ \downarrow \lambda & & \downarrow T \\ \mathbf{M}(B) & \xrightarrow{[\delta_{\Upsilon(\mathbf{M}(B))}]^\lambda} & \mathbf{L}\Upsilon(\mathbf{M}(-)) := \Omega \end{array}$$

From the preceding we conclude that if the evaluation of the localized unit at $\#_M$ is an isomorphism, then the subobject functor **Sub**: $\mathbf{L}^{op} \rightarrow \text{Sets of } \mathbf{L}$ is representable in $\mathbf{L}$ by:

$$\Omega := \mathbf{L}\Upsilon(\mathbf{M}(-))$$

and thus, $\mathbf{L}$ is endowed with a subobject classifier defined by a universal monic quantum homomorphism, $T := \text{True} : 1 \# \Omega$. In the following section, we will provide an *explicit representation of the classifying Boolean modeled quantum event algebra* by exploiting the *functorial tensor product decomposition of the inductive limit*, and we will apply this representation to the problem of measurement in quantum mechanics.

9.6 Truth Valuations in Boolean Localization

Systems

The quantum classifying event algebra Ω is given by the inductive limit in the category of elements of the Boolean frames modeled subobject functor, that is:

$$\Omega := \mathbf{L}\Upsilon(\mathbf{M}(-)) = \text{Colim}\{ \int(\Upsilon(\mathbf{M}(-)), B) \rightarrow B \rightarrow \mathbf{L} \}$$

where the category of elements of $\#(\mathbf{M}(\#))$ is denoted by $\int(\#(\mathbf{M}(\#)), B)$. Its objects are all pairs (B, λ) , where λ is a subobject of $\mathbf{M}(B)$, that is, a monic quantum homomorphism in $\mathbf{M}(B)$. The morphisms of the category of elements of $\#(\mathbf{M}(\#))$ are given by the arrows $(B', \lambda') \rightarrow (B, \lambda)$, namely, they are those morphisms $v: B' \rightarrow B$ of B for which $\lambda * v = \lambda'$, where $\lambda * v$ denotes the pullback of the subobject λ of $\mathbf{M}(B)$ along v , and λ' is a subobject of $\mathbf{M}(B')$.

The inductive limit in the category of elements of the Boolean frames modeled subobject functor can be equivalently represented as a coequalizer of the coproduct; that is, it can be expressed as a tensor product of functors analogous to the construction of section 9.2:

$$\bigsqcup_{v: B' \rightarrow B} \mathbf{M}(B') \xrightleftharpoons[\eta]{\zeta} \bigsqcup_{(B, \lambda)} \mathbf{M}(B) \xrightarrow{\chi} \Upsilon(\mathbf{M}(-)) \otimes_{\mathcal{B}} \mathbf{M}$$

where the second coproduct is over all the pairs (B, λ) with $\lambda \in \#(\mathbf{M}(B))$ of the category of elements, while the first coproduct is over all the morphisms $v: (B', \lambda') \rightarrow (B, \lambda)$ of that category, so that $v: B' \rightarrow B$ and the condition $\lambda * u = \lambda'$ is satisfied.

First, we may interpret the above representation of the inductive limit (in the category of elements of the Boolean frames modeled subobject functor) in the category *Sets*. In this case, the coproduct $\bigsqcup_{(B, \lambda)} \mathbf{M}(B)$ is a coproduct of sets, which is equivalent to the product $\#(\mathbf{M}(B)) \times \mathbf{M}(B)$ for B in B . The coequalizer is thus the definition of the tensor product $\#(\mathbf{M}(\#)) \# \mathbf{M}$ of the set valued functors:

$$\Upsilon(\mathbf{M}(-)) : \mathbf{B}^{op} \rightarrow \mathbf{Sets}, \quad \mathbf{M} : \mathbf{B} \rightarrow \mathbf{Sets}$$

$$\begin{array}{ccc} \bigsqcup_{B, B'} \Upsilon(\mathbf{M}(B)) \times \text{Hom}(\dot{B}, B) \times \mathbf{M}(\dot{B}) & \xrightarrow[\eta]{\zeta} & \\ \xrightarrow[\eta]{\zeta} \bigsqcup_B \Upsilon(\mathbf{M}(B)) \times \mathbf{M}(B) & \xrightarrow{\chi} & \Upsilon(\mathbf{M}(-)) \otimes \mathbf{M}(B) \end{array}$$

According to the preceding diagram for elements $\lambda \in \#(\mathbf{M}(B))$, $v: B' \rightarrow B$ and $q' \in \mathbf{M}(B')$ the following equations hold:

$$\zeta(\lambda, v, q') = (\lambda * v, q'), \quad \eta(\lambda, v, q') = (\lambda, v(q'))$$

symmetric in $\#(\mathbf{M}(B))$ and $\mathbf{M}$. Hence the elements of the set $\#(\mathbf{M}(B)) \#_B \mathbf{M}$ are all of the form $\chi(\lambda, q)$.

This element can be written as:

$$\chi(\lambda, q) = \lambda \otimes q, \quad \lambda \in \Upsilon(\mathbf{M}(B)), q \in \mathbf{M}(B)$$

Thus if we take into account the definitions of ζ and η above, we obtain:

$$[\lambda * v] \otimes q' = \lambda \otimes v(q'), \quad \lambda \in \Upsilon(\mathbf{M}(B)), q' \in \mathbf{M}(B'), v: B' \rightarrow B$$

We conclude that the set $\#(\mathbf{M}(B)) \#_B \mathbf{M}$ is actually the quotient of the set $\mathfrak{u}_B \#(\mathbf{M}(B)) \times \mathbf{M}(B)$ by the equivalence relation generated by the above equations. Furthermore, if we define $\lambda * v = \lambda'$, $v(q') = q$, where λ' is a subobject of $\mathbf{M}(B')$ and $q \in \mathbf{M}(B)$, we obtain the equation:

$$\lambda' \otimes q' = \lambda \otimes q$$

Next, since pullbacks exist in $\mathbf{L}$, we may consider the arrows $h: \mathbf{M}(D) \rightarrow \mathbf{M}(B)$ and $h': \mathbf{M}(D) \rightarrow \mathbf{M}(B')$ and the following pullback diagram in $\mathbf{L}$:

$$\begin{array}{ccc}
 \mathbf{M}(D) & \xrightarrow{h} & \mathbf{M}(B) \\
 \downarrow \hat{h} & & \downarrow \\
 \mathbf{M}(\hat{B}) & \xrightarrow{\quad} & L
 \end{array}$$

such that the following relations are satisfied: $h(d) = q$, $h'(d) = q'$ and $\lambda * h = \lambda' * h'$. Then we obtain:

$$\lambda \otimes q = \lambda \otimes h(d) = [\lambda * h] \otimes d = [\lambda' * h'] \otimes d = \lambda' \otimes h'(d) = \lambda' \otimes q'$$

We may further define:

$$\lambda * h = \lambda' * h' = \tau$$

where τ is a subobject of $\mathbf{M}(D)$. Then, it is obvious that:

$$\lambda \otimes q = \tau \otimes d$$

$$\lambda' \otimes q' = \tau \otimes d$$

It is then clear that we may define a partial order on the set $\#(\mathbf{M}(\#)) \#_{\mathbf{B}} \mathbf{M}$ as follows:

$$\lambda \otimes b \leq \rho \otimes c$$

if and only if there exist quantum algebraic homomorphisms $\beta: \mathbf{M}(D) \rightarrow \mathbf{M}(B)$ and $\#: \mathbf{M}(D) \rightarrow \mathbf{M}(C)$, and some d_1, d_2 in $\mathbf{M}(D)$, such that: $\beta(d_1) = b$, $\#(d_2) = c$, and $\lambda * \beta = \rho * \# = \tau$. Thus we obtain:

$$\lambda \otimes b = \tau \otimes d_1$$

$$\rho \otimes c = \tau \otimes d_2$$

We conclude that:

$$\lambda \otimes b \leq \rho \otimes c$$

if and only if

$$\tau \otimes d_1 \leq \tau \otimes d_2 \Leftrightarrow d_1 \leq d_2$$

The set $\#(\mathbf{M}(\#)) \#_{\mathbf{B}} \mathbf{M}$ may be further endowed with a maximal element which admits the following presentations:

$$\mathbf{1} = \tau \otimes \mathbf{1} := true \quad \forall \tau \in Y(\mathbf{M}(D))$$

$$\mathbf{1} = id_{\mathbf{M}(B)} \otimes b := true \quad \forall b \in \mathbf{M}(B)$$

and an orthocomplementation operator:

$$[\tau \otimes d]^{\hat{a}} = \tau \otimes d^{\hat{a}}$$

Then it is easy to verify that the set $\Omega = \#(\mathbf{M}(\#)) \#_{\mathbf{B}} \mathbf{M}$, endowed with the prescribed operations, is actually a *quantum event algebra* for every Boolean event algebra B in $\mathbf{B}$. Consequently, the elements of the quantum classifying algebra $\Omega = \#(\mathbf{M}(\#)) \#_{\mathbf{B}} \mathbf{M}$ are equivalence classes (*Boolean germs of truth*) represented in tensor product form as follows:

$$[\delta_{Y(\mathbf{M}(B))}]^{\lambda}(b) := \Delta^{\lambda}(b) = \lambda \otimes b$$

where:

$$[\lambda * v] \otimes b' = \lambda \otimes v(b'), \quad \lambda \in Y(\mathbf{M}(B)), b' \in \mathbf{M}(B'), v: B' \rightarrow B, v(b') = b$$

and $[\delta_{\#(\mathbf{M}(B))}]^{\lambda} = \Delta^{\lambda}$ denotes a local Boolean cover of Ω in the Boolean localization system $[\delta_{\#(\mathbf{M}(\#))}]^{\#}$ of Ω using the unit isomorphism.

As a consequence, we obtain the following *criterion of truth valuations with respect to a Boolean localization system* of a quantum event algebra L :

$$[\delta_{Y(\mathbf{M}(B))}]^{\lambda}(b) := \Delta^{\lambda}(b) = \lambda \otimes b = true \quad \text{iff} \quad b \in Image(\lambda)$$

Thus, we have shown explicitly that the category of quantum event algebras L endowed with a Boolean localization system has a *classifying object* Ω , which is defined by the inductive limit in the category of elements of the Boolean frames modeled subobject functor, viz., $\Omega = \#(\mathbf{M}(\#)) \#_{\mathbf{B}} \mathbf{M}$. The elements of the classifying quantum event algebra Ω , that is, the *Boolean germs of truth*, are equivalence classes represented in tensor product form as follows:

$$\Delta^{\lambda}(b) = \lambda \otimes b$$

where:

$$[\lambda * v] \otimes b' = \lambda \otimes v(b'), \quad \lambda \in Y(\mathbf{M}(B)), b' \in \mathbf{M}(B'), v: B' \rightarrow B, v(b') = b$$

and Δ^λ denotes a local Boolean cover of Ω in the Boolean localization system $\Delta^{(\#)}$ of Ω using the unit isomorphism. Thus, we have proved that *truth valuations in quantum mechanics are localized with respect to equivalence classes of compatible Boolean frames belonging to a Boolean localization system of a quantum event algebra*. We require that a Boolean localization system of L in L is closed with respect to Boolean covers, viz., it contains all Boolean frames covering L and satisfying the compatibility conditions. In this way, the logical classifying Boolean germ *true* (equivalence class) used for the valuation of quantum propositions is characterized by:

$$\tau \otimes 1 := \text{true} \quad \forall \tau \in Y(\mathbf{M}(D))$$

$$id_{\mathbf{M}(B)} \otimes b := \text{true} \quad \forall b \in \mathbf{M}(B)$$

$$\lambda \otimes b = \text{true} \quad \text{iff} \quad b \in \text{Image}(\lambda), \quad \forall \lambda \in Y(\mathbf{M}(B))$$

9.7 Boolean Germ Classification of Measurement Events

The classifying quantum event algebra Ω in L , induced by a Boolean localization system, plays a role similar to the role played by the Boolean algebra $2 := \{0,1\}$ in characterizing the logic of propositions referring to the behavior of classical systems. *Thus in the quantum case, subobjects of a quantum event algebra should be depicted in terms of characteristic functions taking values in Ω* . Let us explain the functionality of the logical classifying object Ω according to the criterion of truth valuations by the following diagram:

$$\begin{array}{ccc} \text{Dom}(l * e) & \longrightarrow & K \\ \downarrow l * e = \lambda & & \downarrow l \\ \mathbf{M}(B) & \xrightarrow{e} & L \end{array}$$

where $l: K^\circ L$ is a subobject of a quantum event algebra L , $e: \mathbf{M}(B) \rightarrow L$ is a Boolean cover of L in a Boolean localization system of L , and $l^*e = \lambda$ is the pullback of l along e , that is, the subobject λ of $\mathbf{M}(B)$. According to the truth-value criterion, the characteristic function of the subobject $l: K^\circ L$ of L is specified as an equivalence class of pullbacks of the subobject l along its restrictions on all Boolean covers in a Boolean localization system of L . For each Boolean cover $\mathbf{M}(B)$ of L , $(l^*e)^\# b = \lambda \times b = \text{true}$ if and only if b belongs to the $\text{Image}(\lambda)$, for all λ in $\#(\mathbf{M}(B))$. Thus, for each Boolean cover $\mathbf{M}(B)$ of L , the value $\mathbf{1} = \text{true}$ in Ω is assigned to all those b in $\mathbf{M}(B)$ belonging to the restriction of the subobject $l: K^\circ L$ of L with respect to the subobject λ of $\mathbf{M}(B)$, for all λ . In particular, if the Boolean covers are monomorphisms, each pullback is expressed as the intersection of the subobject l with the corresponding cover in the Boolean localization system.

Conceptually, every quantum event or proposition of a quantum event algebra L is contextualized with respect to all Boolean frames $\mathbf{M}(B)$ belonging to a Boolean localization system of L by means of the operation of restriction. In this way, a *quantum proposition refers to the germ (equivalence class) of all its restricted propositions with respect to all Boolean frames $\mathbf{M}(B)$ belonging to a Boolean localization system of L* . Note that all restricted propositions are *logically compatible* since they belong to a Boolean localization system of L . With respect to each such contextualization we obtain a *contextual truth valuation* of the restricted proposition associated with the corresponding frame $\mathbf{M}(B)$, specified by the rule $\lambda \# b = \text{true}$ if and only if b belongs to the $\text{Image}(\lambda)$, holding for every subobject λ of $\mathbf{M}(B)$, where b represents the restricted quantum proposition with respect to the Boolean frame $\mathbf{M}(B)$. We emphasize that all these contextual truth valuations of some quantum proposition are appropriately related to each other by the formation of equivalence classes *gluing together all its restrictions* with respect to all Boolean frames $\mathbf{M}(B)$ belonging to a Boolean localization system. This is achieved via the logical classifying ‘Boolean germ *true*’ specification constraints in Ω , thus yielding a complete inductive description of reality, defined by: $\tau \# \mathbf{1} := \text{true}$ for all $\tau \in \#(\mathbf{M}(B))$ and $\text{id}_{\mathbf{M}(B)}^\# b := \text{true}$ for all $b \in \mathbf{M}(B)$ and for all Boolean frame domains $\mathbf{M}(B)$ in a Boolean localization system.

Let us now apply the above truth valuation scheme to a typical measurement situation, such as the one occurring in the double-slit optics experiments discussed in chapters 2 and 4. The key measurement in these experiments is the measurement of a quantum system (e.g., a photon) passing through a slit, where a counter on the measuring apparatus will record passage through the slit by clicking (or not clicking if the photon does not pass through). If we denote a Boolean domain preparation context, which contains both the measuring apparatus as well as the system observed by $\mathbf{M}(B)$, then we may form the propositions:

$$\langle c \rangle := [\textit{counter clicks}]$$

$$\langle d \rangle := [\textit{system passes through the slit}]$$

$$\langle c \Rightarrow d \rangle := \langle b \rangle := [\textit{counter clicks} \Rightarrow \textit{system passes through the slit}].$$

Notice that the proposition $\#b\#$ is a *compound proposition* which refers to the *coupling* of the measuring apparatus together with the quantum system in the Boolean frame $\mathbf{M}(B)$. The proposition $\#c \rightarrow d\# := \#b\#$ is assigned the value *true* in Ω , expressing a complete description of the state of affairs. More precisely, with respect to the Boolean logical frame of $\mathbf{M}(B)$, or more concretely with respect to $id_{\mathbf{M}(B)}$, we have that $id_{\mathbf{M}(B)} \#b = \textit{true}$. We notice that the above truth valuation *does not* suffice in order to infer that the proposition $\#d\#$ is true. In order to infer the above, we need to use the Boolean frame $\mathbf{M}(C)$ which contains only the measuring apparatus, which is a *subobject* of the Boolean frame $\mathbf{M}(B)$, viz., $\lambda : \mathbf{M}(C) \# \mathbf{M}(B)$. Then, we obtain that: $id_{\mathbf{M}(C)} \#c = \textit{true}$ and $\lambda \# \xi = \textit{true}$ if and only if ξ belongs to the *Image* (λ) in $\mathbf{M}(B)$. The Boolean contexts $\mathbf{M}(C)$ and $\mathbf{M}(B)$ induce the formation of a *Boolean localization system* for the evaluation of $\#d\#$ and thus they are *glued* compatibly together. Hence, we deduce that $\lambda \#b = \textit{true}$ because b belongs to the *Image*(λ) in $\mathbf{M}(B)$. In particular b in $\mathbf{M}(B)$ is the image of c in $\mathbf{M}(C) \circ \mathbf{M}(B)$ and the truth of $\#d\#$ is *indirectly* inferred by the measurement event through the *Boolean germtrue* (equivalence class) defined by:

$$id_{\mathbf{M}(C)} \otimes c = \lambda \otimes b = id_{\mathbf{M}(B)} \otimes b = \textit{true}$$

Notes

1. V. Karakostas, “Non-separability, “Potentiality and the Context-Dependence of Quantum Objects,” *Journal for General Philosophy of Science* 38 (2007).
2. P.T. Johnstone, *Stone Spaces* (Cambridge: Cambridge University Press, 1986).
3. Elias Zafiris, “Topos-Theoretic Classification of Quantum Events Structures in Terms of Boolean Reference Frames,” *International Journal of Geometric Methods in Modern Physics* 3, no. 8 (2006).